```
1  x := H(x);
2  x := Z(x);
3  assert x == |−⟩;  A1
4  y := H(y);
5  assert y == |+⟩;  A2
6  x,y := CNOT(x,y);
7  assert y == |+⟩;  A3
8  x := H(x);
9  y := H(y);
10 measure x;
11 measure y;
```

Two Naive Strategies

one *- round /* ***three*** *- ancillas*

one *- ancilla /* ***three*** *- rounds*

program executions **# ancillas per execution**

$$E \times S = O(n)$$

Can we achieve a better complexity ?

Task	Single-Round ($E = 1$)	Disjoint Multi-Round ($E \geq 1$)	General Multi-Round ($E \geq 1$)
ExistFail whether **exists** a failure	$S = \Theta(\log n)$	$S = \Theta(\log(1 + \frac{n}{E}))$	$S = \Theta(\frac{\log n}{E}) \quad \forall E \in O(\frac{\log n}{\log\log n})$
FirstFail find the **first** failure	$S = \Theta(\log n)$	$S = \Theta(\log(1 + \frac{n}{E}))$	$S = \Theta(\log(1 + \frac{n}{E}))$
ListAll output the **full** outcomes	$S = \Theta(n)$	$S = \Theta(\frac{n}{E})$	$S = \Theta(\frac{n}{E})$

- complexity hinges on the information target: *logarithmic* vs. *linear*
- one can pay more time for less space: different *trade-offs*